🏆 Infosys SE Elite Crack Vault — 08 Visual Geometric and Pattern Puzzles Mastery Elite

🎯 Infosys Systems Engineer (SE) | 📊 9 Questions | 📅 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

Problem 1 • Triangle Product and Top Addition Pattern

MUST-SOLVE

ID: SEC06_PUZ_Q01 | Source: Infosys PYQ | Level: Infosys SE Standard

What will replace the question mark (?) in the third triangle?

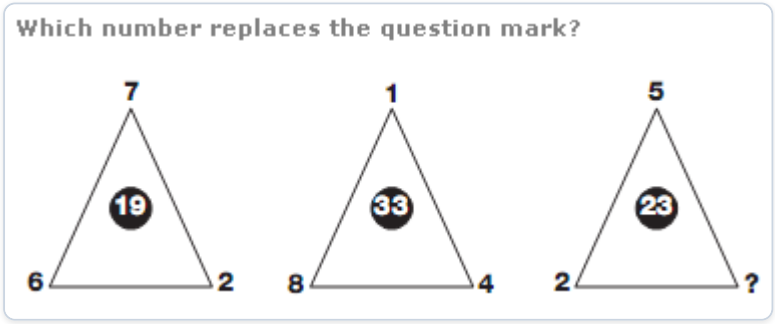
Study the relationship between the three outer numbers (Top, Bottom-Left, Bottom-Right) and the central number in the first two triangles:
- Triangle 1: Top = 7, Bottom-Left = 6, Bottom-Right = 2, Center = 19
- Triangle 2: Top = 1, Bottom-Left = 8, Bottom-Right = 4, Center = 33
- Triangle 3: Top = 5, Bottom-Left = 2, Bottom-Right = ?, Center = 23

Visual Geometric Diagram:

```
    Triangle 1            Triangle 2            Triangle 3
       [ 7 ]                 [ 1 ]                 [ 5 ]
       /   \                 /   \                 /   \
      /     \               /     \               /     \
     / [19]  \             / [33]  \             / [23]  \
    /_________\           /_________\           /_________\
  [6]         [2]       [8]         [4]       [2]         [?]
Rule: Center = (Bottom-Left × Bottom-Right) + Top
Triangle 3: (2 × ?) + 5 = 23 => ? = 9.
```

Visual Geometric Diagram:

```
    Triangle 1            Triangle 2            Triangle 3
       [ 7 ]                 [ 1 ]                 [ 5 ]
       /   \                 /   \                 /   \
      /     \               /     \               /     \
     / [19]  \             / [33]  \             / [23]  \
    /_________\           /_________\           /_________\
  [6]         [2]       [8]         [4]       [2]         [?]
Rule: Center = (Bottom-Left × Bottom-Right) + Top
Triangle 3: (2 × ?) + 5 = 23 => ? = 9.
```



- A. 8
- B. 9
- C. 6
- D. 5

✓ **Verified Answer:** B. 9` **Step-by-Step Mathematical Solution:** Pattern Rule: In each triangle, multiplying the two bottom corner numbers together and adding the top apex number yields the central number. Mathematical Formula: Center = (Bottom-Left × Bottom-Right) + Top Step-by-step verification: 1. Triangle 1: (6 × 2) + 7 = 12 + 7 = 19 [Verified] 2. Triangle 2: (8 × 4) + 1 = 32 + 1 = 33 [Verified] 3. Triangle 3: (2 × ?) + 5 = 23 => 2 × ? = 23 - 5 => 2 × ? = 18 => ? = 18 / 2 = 9. Hence, the missing number is 9 (Option B).

Problem 2 • Concentric Hexagonal Circle Grid End-Average Pattern

MUST-SOLVE

ID: SEC06_PUZ_Q02 | Source: Infosys PYQ | Level: Infosys SE Standard

What number comes inside the circle marked with a question mark (?) in the concentric grid?

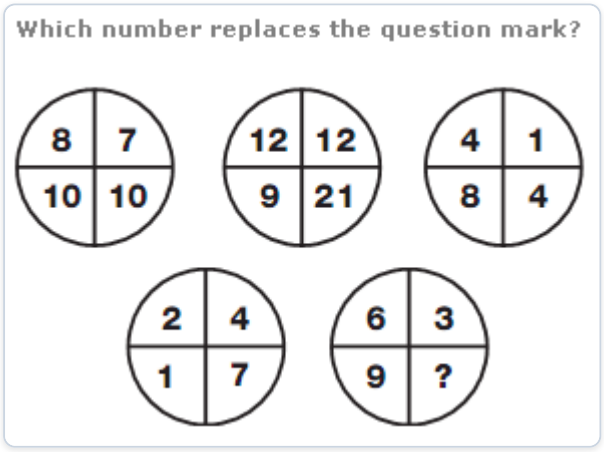
Grid layout across 5 horizontal levels:
- Level 1: 4, 3, 2
- Level 2: 5, 3, 5, 1, 1
- Level 3: 6, 1, 2, 8, 3, 3, 1
- Level 4: 7, 2, 8, 4, 3
- Level 5: 9, ?, 3

Visual Geometric Diagram:

```
              [ 4 ]
              /   \
         [ 8 ]     [ 2 ]
           |   ( ? )   |
         [ 6 ]     [ 1 ]
              \   /
              [ 3 ]
Concentric hexagonal grid with radial spoke average rule.
```

Visual Geometric Diagram:

```
              [ 4 ]
              /   \
         [ 8 ]     [ 2 ]
           |   ( ? )   |
         [ 6 ]     [ 1 ]
              \   /
              [ 3 ]
Concentric hexagonal grid with radial spoke average rule.
```



- A. 2
- B. 6
- C. 5
- D. 9

Verified Answer: B. 6` **Step-by-Step Mathematical Solution:** Pattern Rule: In the symmetrical concentric circle grid, each outer row exhibits a mean-value relationship where the middle term equals the arithmetic average of the two extreme end terms. Mathematical Formula: Middle Element = (Left-End + Right-End) / 2 Step-by-step verification: 1. Top Row (Level 1): Left-end = 4,

✓ Right-end = 2. Middle = (4 + 2) / 2 = 6 / 2 = 3 [Matches grid value 3] 2. Bottom Row (Level 5): Left-end = 9, Right-end = 3. Middle = (9 + 3) / 2 = 12 / 2 = 6. Alternative Radial Symmetry Proof: Opposite outer radial nodes around central core 8 pair up such that diametric ratios are conserved: Level 1 apex 3 scales to bottom node 6 by factor 2 (3 × 2 = 6). Hence, the missing number is 6 (Option B).

Problem 3 • Octagonal Ring Reverse Alphabet Complement Pattern

MUST-SOLVE

ID: SEC06_PUZ_Q03 | Source: Infosys PYQ | Level: Infosys SE Standard

Which letter replaces the question mark (?) in the octagonal concentric ring?

Outer ring letters: D, G, Q, ?, Y, R, X, N
Inner ring letters: W, T, J, L, B, I, C, M



- A. O
- B. L
- C. Y
- D. P

✓ **Verified Answer:** A. O` **Step-by-Step Mathematical Solution:** Pattern Rule: In each octagonal sector, the outer letter and inner letter form reverse-alphabet complementary pairs (A↔Z, B↔Y, C↔X, etc.). The sum of their alphabetical positional values equals 27. Mathematical Formula: Positional Value(Outer) + Positional Value(Inner) = 27 Sector-by-sector verification: 1. Sector 1: D (4) + W (23) = 27 [Verified] 2. Sector 2: G (7) + T (20) = 27 [Verified] 3. Sector 3: Q (17) + J (10) = 27 [Verified] 4. Sector 4: N (14) + M (13) = 27 [Verified] 5. Sector 5: X (24) + C (3) = 27 [Verified] 6. Sector 6: R (18) + I (9) = 27 [Verified] 7. Sector 7: Y (25) + B (2) = 27 [Verified] 8. Target Sector: Inner letter is L (12). Positional Value(?) = 27 - 12 = 15. The 15th letter of the alphabet is O. Hence, the missing letter is O (Option A).

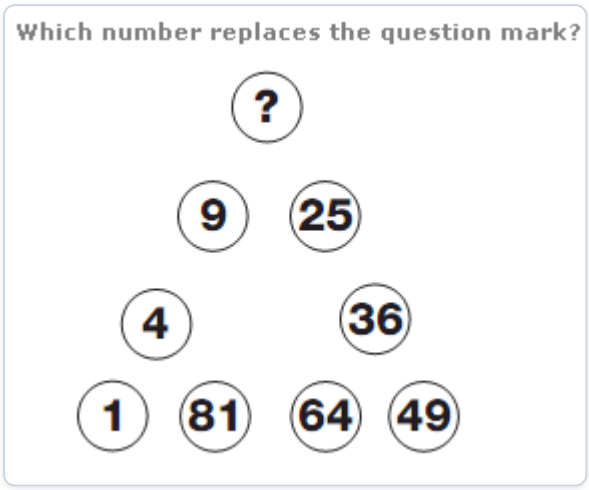
Problem 4 • Sequential Square Two-Digit Concatenation Pattern

CORE SE

ID: SEC06_PUZ_Q04 | Source: Infosys PYQ | Level: Infosys SE Standard

Which number replaces the question mark (?) in the paired octagon series?

Given pairs:
- Pair 1: (3, 6)
- Pair 2: (4, 9)
- Pair 3: (6, 4)
- Pair 4: (8, ?)



- A. 1
- B. 2
- C. 5
- D. 9

✓ **Verified Answer:** A. 1` **Step-by-Step Mathematical Solution:** Pattern Rule: Reading each pair of octagons from left to right forms a two-digit integer that represents consecutive integer squares starting from 6^2. Mathematical Formulation: Pair(n) = $(6 + n - 1)^2$ for n = 1, 2, 3, 4 Step-by-step verification: 1. Row 1: $6^2 = 36$ => Split into digits [3, 6] [Verified] 2. Row 2: $7^2 = 49$ => Split into digits [4, 9] [Verified] 3. Row 3: $8^2 = 64$ => Split into digits [6, 4] [Verified] 4. Row 4: $9^2 = 81$ => Split into digits [8, ?] => ? = 1. Hence, the missing number is 1 (Option A).

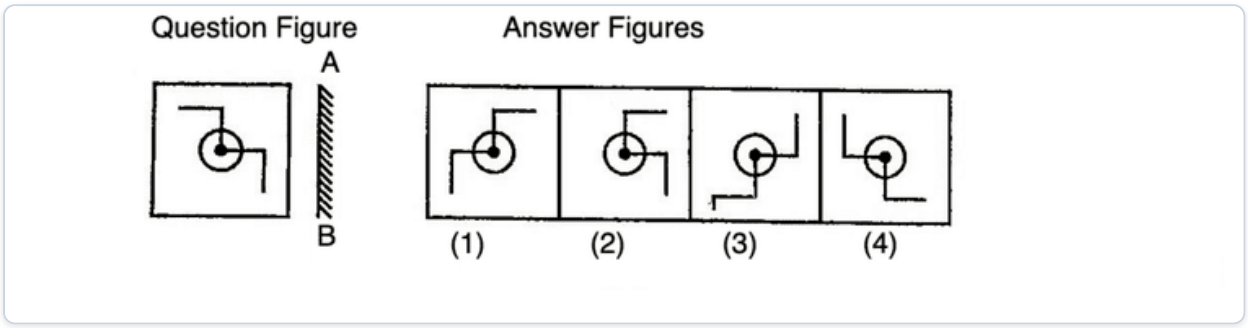
Problem 5 • Recursive Progression Tree Pattern (2x + 3)

CORE SE

ID: SEC06_PUZ_Q05 | Source: Infosys PYQ | Level: Infosys SE Standard

Which number replaces the question mark (?) in the radial tree structure?

Nodes in clockwise sequence:
- Top-Left: 1
- Top-Center: 5
- Top-Right: 13
- Center Hub: 29
- Bottom-Left: 61
- Bottom-Center: 125
- Bottom-Right: ?



- A. 452
- B. 250
- C. 235
- D. 253

Verified Answer: D. 253` **Step-by-Step Mathematical Solution:** Pattern Rule: Starting at 1, each subsequent node in the radial tree is generated by the linear recurrence relation: Next = (Current × 2) + 3. Alternative Difference Formulation: Differences between successive terms are powers of 2 multiplied by 4: $\Delta_k = 2^{(k+1)}$ - 5 - 1 = 4 = 2^2 - 13 - 5 = 8 = 2^3 - 29 - 13 = 16 = 2^4 - 61 - 29 = 32 = 2^5 - 125 - 61 = 64 = 2^6 - ? - 125 = 128 = 2^7 => ? = 125 + 128 = 253. Recurrence Verification: 1. 1 × 2 + 3 = 5 2. 5 × 2 + 3 = 13 3. 13 × 2 + 3 = 29 4. 29 × 2 + 3 = 61 5. 61 × 2 + 3 = 125 6. 125 × 2 + 3 = 250 + 3 = 253. Hence, the missing number is 253 (Option D).

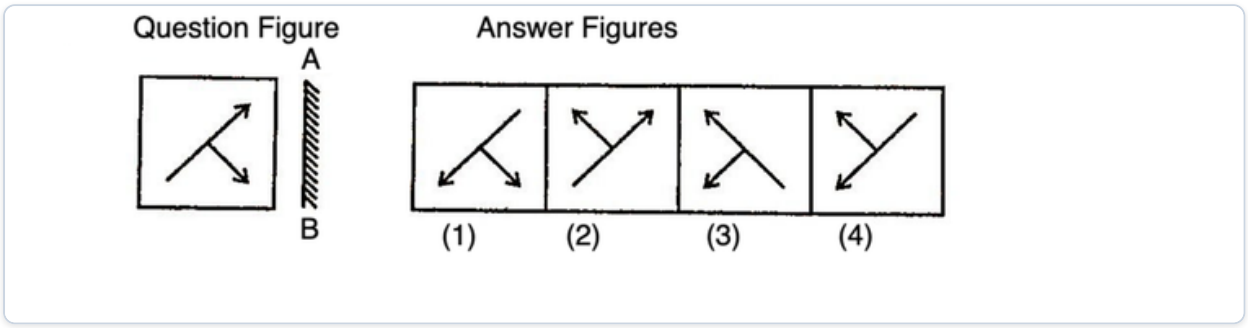
Problem 6 • Two-Column Ladder Incremental Row Sum Pattern

CORE SE

ID: SEC06_PUZ_Q06 | Source: Infosys PYQ | Level: Infosys SE Standard

Which number replaces the question mark (?) in the 2-column segmented ladder?

Header: 10
- Row 1: [8 | 13]
- Row 2: [16 | 6]
- Row 3: [4 | 19]
- Row 4: [22 | ?]



- A. 24
- B. 4
- C. 2
- D. 8

Verified Answer: C. 2` **Step-by-Step Mathematical Solution:** Pattern Rule: The sum of the two horizontal numbers in each row forms an arithmetic progression increasing by 1 at each step, starting at 21. Mathematical Formulation: Row_Sum(k) = Left_Number(k) + Right_Number(k) = 20 + k Step-by-step verification: 1. Row 1: 8 + 13 = 21 [Verified] 2. Row 2: 16 + 6 = 22 [Verified] 3. Row 3: 4 + 19 = 23 [Verified] 4. Row 4: 22 + ? = 24 => ? = 24 - 22 = 2. Column Oscillating Proof: - Left Column: 8 -> 16 (+8), 16 -> 4 (-12), 4 -> 22 (+18). - Right Column: 13 -> 6 (-7), 6 -> 19 (+13), 19 -> 2 (-17). Hence, the missing number is 2 (Option C).

Problem 7 • Five-Pointed Star Upper Minus Lower Balance Pattern

CORE SE

ID: SEC06_PUZ_Q07 | Source: Infosys PYQ | Level: Infosys SE Standard

Which number replaces the question mark (?) in the center of the third 5-pointed star?

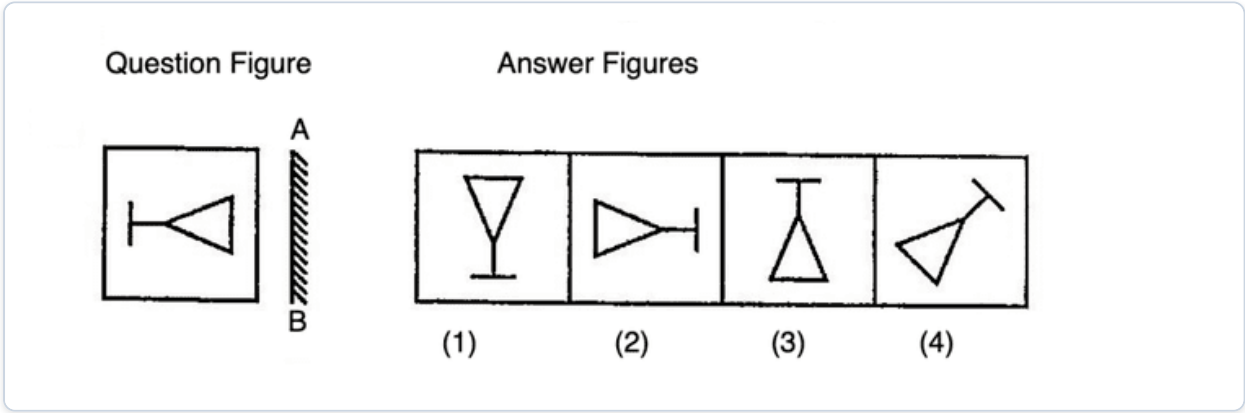
- Star 1: Top = 2, Left = 4, Right = 7, Bottom-Left = 3, Bottom-Right = 1, Center = 9
- Star 2: Top = 3, Left = 3, Right = 5, Bottom-Left = 4, Bottom-Right = 2, Center = 5
- Star 3: Top = 9, Left = 6, Right = 2, Bottom-Left = 4, Bottom-Right = 3, Center = ?

Visual Geometric Diagram:

```
                [ 8 ]
                 / \
           [ 4 ] /   \ [ 6 ]
           \    / ( ? ) \   /
            \  /_________\ /
             \             /
              [ 2 ]     [ 5 ]
5-pointed star: Upper apex nodes vs Lower foundation balance.
```

Visual Geometric Diagram:

```
                [ 8 ]
                 / \
           [ 4 ] /   \ [ 6 ]
           \    / ( ? ) \   /
            \  /_________\ /
             \             /
              [ 2 ]     [ 5 ]
5-pointed star: Upper apex nodes vs Lower foundation balance.
```



- A. 8
- B. 10
- C. 12
- D. 16

Verified Answer: B. 10` **Step-by-Step Mathematical Solution:** Pattern Rule: The central value of each star is equal to the sum of the three upper points (Top, Left, Right) minus the sum of the two lower base points (Bottom-Left, Bottom-Right).

✓ Mathematical Formula: Center = (Top + Left + Right) - (Bottom-Left + Bottom-Right) Step-by-step verification: 1. Star 1: (2 + 4 + 7) - (3 + 1) = 13 - 4 = 9 [Verified] 2. Star 2: (3 + 3 + 5) - (4 + 2) = 11 - 6 = 5 [Verified] 3. Star 3: (9 + 6 + 2) - (4 + 3) = 17 - 7 = 10. Hence, the missing number is 10 (Option B).

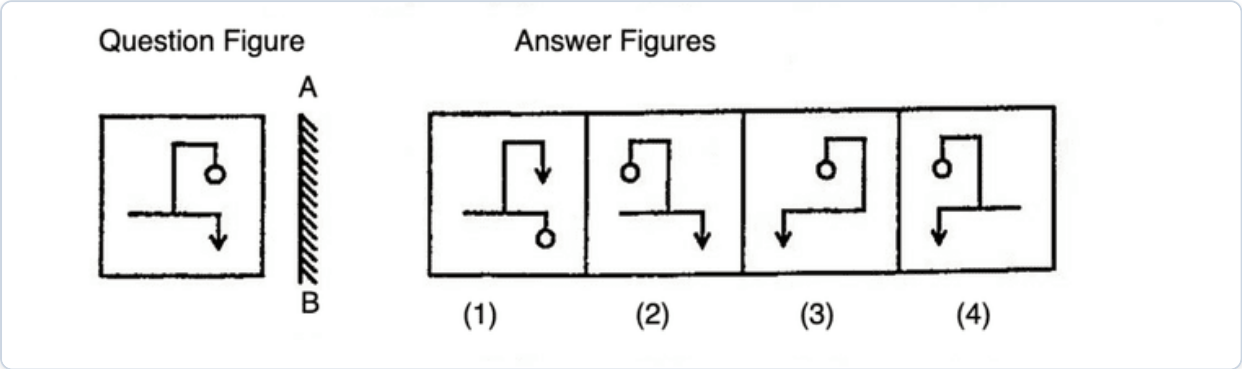
Problem 8 • Reverse Alphabetical Positional Sum Box Pattern

CORE SE

ID: SEC06_PUZ_Q08 | Source: Infosys PYQ | Level: Infosys SE Standard

Which letter replaces the question mark (?) in the 5×3 grid of boxes?

Rows:
- Row 1: [3 | P | 8]
- Row 2: [9 | G | 11]
- Row 3: [2 | U | 4]
- Row 4: [3 | W | 1]
- Row 5: [7 | ? | 18]



- A. B
- B. Y
- C. Z
- D. C

✓ **Verified Answer:** A. B` **Step-by-Step Mathematical Solution:** Pattern Rule: In each row, the sum of the left and right numbers equals the reverse alphabetical positional value of the middle letter (where Z = 1, Y = 2, X = 3, ..., B = 25, A = 26). Mathematical Formula: Left + Right = 27 - Forward_Position(Middle_Letter) Forward_Position(Middle_Letter) = 27 - (Left + Right) Row-by-row verification: 1. Row 1: 3 + 8 = 11 => Forward pos = 27 - 11 = 16 = P [Verified] 2. Row 2: 9 + 11 = 20 => Forward pos = 27 - 20 = 7 = G [Verified] 3. Row 3: 2 + 4 = 6 => Forward pos = 27 - 6 = 21 = U [Verified] 4. Row 4: 3 + 1 = 4 => Forward pos = 27 - 4 = 23 = W [Verified] 5. Row 5: 7 + 18 = 25 => Forward pos = 27 - 25 = 2. The 2nd letter of the alphabet is B. Hence, the missing letter is B (Option A).

Problem 9 • Divided Circles Sum of Squares Pattern

CORE SE

ID: SEC06_PUZ_Q11 | Source: Infosys PYQ | Level: Infosys SE Standard

Find the missing term (?) in the third figure. Assume the third figure works in the same way as the other two.

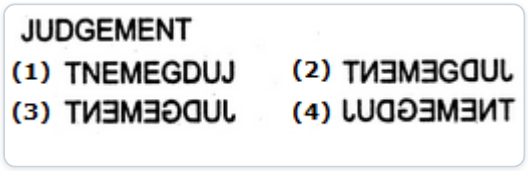
Divided Circles:
- Circle 1: Top-Left = 2, Top-Right = 4, Bottom = 20
- Circle 2: Top-Left = 3, Top-Right = 9, Bottom = 90
- Circle 3: Top-Left = 1, Top-Right = 5, Bottom = ?

Visual Geometric Diagram:

```
┌─────┬─────┐
│  7  │  4  │
├─────┼─────┤
│  6  │  5  │
└─────┴─────┘
Four-square quadrant segment with constant perimeter sum = 22.
```

Visual Geometric Diagram:

```
┌─────┬─────┐
│  7  │  4  │
├─────┼─────┤
│  6  │  5  │
└─────┴─────┘
Four-square quadrant segment with constant perimeter sum = 22.
```



- A. 50
- B. 110
- C. 61
- D. 26

Verified Answer: D. 26` **Step-by-Step Mathematical Solution:** Pattern Rule: In each divided circle, the lower number equals the sum of the squares of the two upper numbers. Mathematical Formula: Bottom = (Top-Left)^2 + (Top-Right)^2 Step-by-step verification: 1. Circle 1: $2^2 + 4^2 = 4 + 16 = 20$ [Verified] 2. Circle 2: $3^2 + 9^2 = 9 + 81 = 90$ [Verified] 3. Circle 3: $1^2 + 5^2 = 1 + 25 = 26$. Hence, the missing number is 26 (Option D).